

RANOpt AI: An Explainable AI-Assisted Interactive Decision Platform for 5G/6G RAN Engineering

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Abstract—The rapid evolution from fifth-generation (5G) mobile systems toward AI-native sixth-generation (6G) networks has significantly increased the complexity of Radio Access Network (RAN) engineering. Engineers must simultaneously evaluate coverage, capacity, spectral efficiency, and energy trade-offs across increasingly heterogeneous deployment scenarios, often without a rapid, transparent way to explore alternative configurations before committing to detailed planning. This paper presents RANOpt AI, a browser-based, AI-assisted interactive decision-support platform for 5G/6G RAN engineering. The platform integrates standards-aligned deterministic engineering models – covering propagation, link budget, SINR estimation, and MAC scheduling – with an AI Decision Engine that presently combines explainable rule-based logic with a Proximal Policy Optimization (PPO)-based reinforcement learning policy already deployed in the browser for scheduler-optimization inference, while continued offline training and pretraining refinement of that policy against broader operating conditions and operator data remain ongoing work. A layered validation framework, comprising configuration checks, golden engineering scenarios, and a diagnostic trace mode, supports reproducible and auditable engineering studies. We describe the system architecture, the underlying wireless engineering models, the reinforcement learning framework deployed for scheduler-optimization inference, and the validation methodology, and we outline a roadmap toward a multi-cell, AI-native Network Digital Twin as a planned future upgrade to the platform.

Index Terms—5G, 6G, Radio Access Network, reinforcement learning, Proximal Policy Optimization, decision support, digital twin, explainable AI, MAC scheduling

I. INTRODUCTION

Modern cellular networks must simultaneously optimize coverage, capacity, spectral efficiency, energy consumption, latency, reliability, user mobility, and quality of experience across increasingly heterogeneous deployment environments. Emerging technologies such as massive MIMO, carrier aggregation, dynamic spectrum utilization, AI-native architectures, and integrated sensing further expand the number of interacting design parameters that engineers must evaluate. Traditional manual engineering approaches become progressively more challenging as this complexity grows.

While sophisticated commercial planning and optimization tools exist, many engineering studies, technology evaluations, and “what-if” analyses still require significant expertise, lengthy configuration procedures, or large-scale simulation environments. There is a practical need for a rapid, intuitive way to evaluate alternative radio configurations before committing to detailed network planning or field validation.

Artificial Intelligence (AI), and Reinforcement Learning (RL) in particular, offers the potential to assist engineers in identifying effective operating points and recommending parameter adjustments. In this work, AI is treated as an *engineering decision-support assistant* rather than a replacement for engineering judgment: the platform preserves full transparency of every assumption and recommendation, and final engineering decisions remain with the user.

This paper presents **RANOpt AI**, an AI-assisted interactive 5G/6G RAN decision platform. Its objective is to provide a browser-based environment that combines standards-aligned engineering models, interactive visualization, natural-language interaction, and AI-assisted decision support to enable rapid evaluation of alternative radio configurations for a representative cell operating within a specified coverage area and deployment morphology. The platform is not intended to replace commercial radio planning tools, operational support systems, or self-organizing network (SON) platforms; rather, it complements them as a fast, explainable, and standards-aligned engineering decision layer.

The remainder of this paper is organized as follows. Section II presents the system architecture. Section III describes the underlying wireless engineering models. Section IV presents the AI Decision Engine and the reinforcement learning framework, including its application to MAC scheduling. Section V describes the validation methodology. Section VI outlines the future evolution of the platform, including the planned area-wide digital twin upgrade. Section VII concludes the paper.

II. SYSTEM ARCHITECTURE

RANOpt AI integrates browser-based interaction, deterministic wireless engineering calculations, AI-assisted recommendations, KPI computation, and visualization within a single modular workflow. The architecture is organized into a set of cooperating functional subsystems:

- **Browser-Based User Interface:** provides graphical parameter selection, natural-language interaction, and results presentation, without requiring dedicated software installation.
- **Engineering Calculation Engine:** performs deterministic, standards-aligned wireless engineering calculations covering propagation, link budget, coverage, and capacity.
- **AI Decision Engine:** generates engineering recommendations, currently combining explainable rule-based logic

with PPO-based reinforcement-learning policy inference; continued offline training of that policy is ongoing work.

- **KPI Computation Engine:** derives representative performance metrics, including coverage, throughput, spectral efficiency, and resource utilization.
- **Visualization Engine:** presents numerical results, comparative analyses, and AI recommendations through an interactive dashboard.

User inputs initiate deterministic engineering calculations; the AI engine analyzes the resulting network state and recommends optimization opportunities; KPIs are computed; and results are presented through interactive visualizations. Throughout this process, the engineer retains complete control over every configuration parameter and optimization decision. The workflow is iterative: engineers repeatedly adjust parameters, evaluate scenarios, and compare results, supporting rapid engineering exploration.

Although the current implementation is intentionally cell-centric – modeling an individual radio cell operating within a specified coverage area and deployment morphology – the modular architecture is designed to accommodate future expansion, including multi-cell modeling, realistic mobility, and Digital Twin functionality, with minimal modification to the existing system.

A. Interaction Modes

To accommodate users with different levels of technical expertise, the interface supports two complementary modes. *Normal Mode* presents frequently used parameters through a simplified interface intended for engineering studies, demonstrations, education, and technology evaluation. *Expert Mode* exposes the complete set of engineering parameters, detailed KPI calculations, validation tools, and diagnostic trace information, intended for experienced engineers requiring fine control over simulation assumptions.

The browser architecture also provides the foundation for *conversational engineering*: while graphical interaction is primary today, the platform is designed to support natural-language and, in future releases, voice-based commands (e.g., “compare AI on and AI off performance”) without altering the underlying deterministic engineering workflow.

III. WIRELESS ENGINEERING MODELS

The reliability of any RAN decision-support platform depends fundamentally on the fidelity of its underlying engineering models. RANOpt AI adopts a model-based engineering approach that combines deterministic wireless communication models with AI-assisted optimization, prioritizing rapid, repeatable “what-if” analysis over full event-driven network simulation.

A. Propagation and Link Budget

The platform implements representative large-scale propagation models for Dense Urban, Urban, and Suburban deployment morphologies, consistent with the framework of 3GPP

TR 38.901. The simplified downlink link budget is expressed as

$$P_{RX} = P_{TX} + G_{TX} + G_{RX} - L_{Path} - L_{Misc}, \quad (1)$$

where P_{TX} is the gNB transmit power, G_{TX} and G_{RX} are the transmit and receive antenna gains, L_{Path} is the propagation path loss, and L_{Misc} represents feeder, penetration, and implementation losses. Downlink and uplink link budgets are evaluated independently, using UE and gNB receiver noise figures respectively.

B. Noise Floor and SINR

The receiver noise floor is calculated as

$$\text{Noise Floor (dBm)} = -174 + 10 \log_{10}(B) + NF, \quad (2)$$

where B is the receiver bandwidth in Hz and NF is the receiver Noise Figure in dB. The resulting Signal-to-Interference-plus-Noise Ratio (SINR) is computed as

$$\text{SINR (dB)} = P_{RX} - 10 \log_{10}(P_I + P_N), \quad (3)$$

combining received signal power with interference and thermal noise. SINR drives CQI determination, MCS selection, spectral efficiency estimation, and ultimately throughput and coverage.

C. Channel Abstraction

Rather than generating full 3GPP Clustered Delay Line (CDL) channel realizations – which would require explicit spatial MIMO channel matrices, angular statistics, and time-varying fading – the platform represents CDL-A through CDL-E as calibrated implementation-efficiency factors reflecting the relative impact of representative LOS/NLOS propagation environments on achievable spectral efficiency. This system-level abstraction is intentional, trading physical-layer fidelity for computational efficiency suitable for interactive, browser-based AI optimization studies.

D. Spectral Efficiency

Spectral efficiency, η (bits/s/Hz), links physical-layer radio quality to higher-layer throughput. While the theoretical Shannon bound is $\eta_{max} = \log_2(1 + \text{SINR})$, practical spectral efficiency in RANOpt AI is derived from SINR-dependent MCS selection combined with CDL channel-efficiency factors, MIMO spatial-multiplexing gain, carrier-aggregation gain, and scheduler efficiency.

E. MAC Scheduler

The MAC scheduler determines which UEs are served and how radio resources are allocated among them, directly influencing cell throughput, user throughput, latency, and fairness. The primary scheduler implemented is Proportional Fair (PF), with scheduling metric

$$PF_i = \frac{R_i^{inst}}{R_i^{avg}}, \quad (4)$$

where R_i^{inst} is the instantaneous achievable rate for user i (from CQI/MCS) and R_i^{avg} is that user’s historical average

throughput. The architecture also accommodates Round Robin and Maximum Throughput scheduling, and is designed for future AI-assisted scheduling policies.

IV. AI DECISION ENGINE AND REINFORCEMENT LEARNING FRAMEWORK

The AI Decision Engine evaluates the network state produced by the Engineering Calculation Engine – coverage, SINR, CQI, spectral efficiency, and scheduler utilization – and recommends optimization actions. Consistent with a human-in-the-loop design, it does not act autonomously: every recommendation is presented with its supporting rationale, expected KPI impact, and a confidence indicator, and the engineer decides whether to accept it.

A. Current Rule-Based Implementation

The present AI implementation is a rule-based expert system in which engineering knowledge is encoded as logical rules – for example, recommending increased bandwidth when spectrum-limited, higher-order MIMO where conditions allow, or carrier aggregation when component carriers are available. Because every recommendation traces to an explicit rule, this approach is fully explainable and auditable.

B. Rule-Based and PPO-Based Reinforcement Learning

The AI Decision Engine presently combines two complementary mechanisms. Rule-based recommendations, described above, remain available as a fully transparent baseline. In parallel, a Proximal Policy Optimization (PPO) policy – an actor-critic reinforcement learning algorithm selected for its stable convergence and effective handling of large decision spaces – is already deployed in the browser application for scheduler-optimization *inference*. The RL formulation follows the standard Markov Decision Process:

- **State:** carrier configuration, MIMO mode, user loading, coverage, and SINR/CQI distribution;
- **Action:** scheduler parameter adjustment, carrier-aggregation selection, MIMO mode optimization, and resource-allocation strategy;
- **Reward:** a multi-objective signal combining throughput, cell-edge throughput, spectral efficiency, coverage, fairness, and latency;
- **Policy:** updated via PPO to maximize long-term, multi-objective network performance.

A key architectural feature is the separation of training from inference. The browser application loads a pre-trained policy and performs inference only, generating recommendations without incurring the cost of online training; this keeps the platform fast, deterministic, and fully browser-portable. What remains ongoing work is the offline *pretraining* pipeline itself – continued training of the policy across a wider range of representative network scenarios and, where available, anonymized carrier data on traffic loading and spectrum configuration – so that only progressively better-trained policy weights, never raw carrier data, are distributed with the application. In other words, PPO inference is a

present capability of the platform; broadening and refining the policy through additional offline pretraining is the part of the framework still evolving.

C. Explainability

Irrespective of whether a recommendation originates from rule-based logic or the deployed PPO policy, the AI Decision Engine is designed to expose the triggering network condition, the recommended action, the expected KPI improvement, associated trade-offs, and a confidence level. For the rule-based path this traceability is direct, since each recommendation maps to an explicit rule. For the PPO path, where recommendations arise from learned policy weights rather than legible rules, achieving the same standard of transparency is a distinct and ongoing design challenge, motivating post-hoc explanation techniques such as attributing a given action to the most influential state variables. This design choice reflects the broader recognition that explainability is essential for improving trust between human operators and machine-driven decisions in AI-native 6G networks [6], and for practical AI deployment in Open RAN more generally.

V. VALIDATION METHODOLOGY

Because RANOpt AI is intended to support engineering decisions rather than merely display calculated values, every result must be traceable to a defined configuration and a consistent calculation sequence. The validation framework follows four principles: configuration validity, calculation traceability, engineering plausibility, and repeatability.

A *Validation Engine* checks band/duplex consistency, bandwidth constraints, carrier-aggregation validity, MIMO/UE-capability consistency, and KPI range and monotonicity (e.g., non-negative throughput, coverage bounded within [0, 100]%). A *Quick Validation Test* provides an immediate pass/fail readiness check suitable for repeated use during interactive studies, while a *Detailed Validation Test* performs a broader review of the full calculation chain, including link-budget arithmetic, SINR-to-CQI mapping, and scheduler resource conservation. *Trace Mode* exposes intermediate quantities – path loss, SINR, CQI, spectral efficiency, and resource allocation – for expert-level diagnostic review.

A set of *golden engineering scenarios* – spanning coverage-limited low-band FDD, balanced mid-band FDD, capacity-oriented mid-band TDD, FDD-plus-TDD layered, and AI-assisted carrier-aggregation configurations – provides repeatable reference cases for regression testing, protecting against unintended changes in calculated behavior as the software evolves. AI-assisted performance is always evaluated through controlled comparison against an unchanged baseline configuration, with both baseline and optimized values retained, so that improvements are not misattributed to AI when they instead result from a simultaneous change in bandwidth, spectrum, or traffic assumptions.

VI. FUTURE WORK

RANOpt AI has been deliberately designed as an evolutionary platform, adopting a phased development strategy

rather than attempting to implement every aspect of future 6G network intelligence in its initial release. Planned directions include:

- **Continued PPO pretraining:** broadening the offline training pipeline described in Section IV – across a wider range of morphologies, traffic conditions, and, where available, anonymized carrier data – to progressively refine the PPO policy already deployed for browser-based inference, together with development of post-hoc explanation methods for its recommendations.
- **Multi-cell modeling:** extending the engineering scope from a single representative cell to multiple interacting cells within a geographical service area, introducing inter-cell interference, coordinated scheduling, and user mobility.
- **AI-native Network Digital Twin:** the longer-term goal is an AI-native RAN Digital Twin in which a high-fidelity virtual representation of a real wireless network continuously mirrors its operational state, enabling engineers to evaluate software updates, AI algorithms, and optimization strategies in a safe virtual environment prior to live deployment.
- **Area/morphology-based digital twin with carrier data:** as a distinct and separate future upgrade *not part of the current version* of RANOpt AI, work is in progress to extend the platform toward an area- and morphology-based digital twin informed by either carrier/operator network data or synthetic data. This extension is architecturally consistent with, but independent of, the single-cell decision-support capability described in this paper.
- **Conversational and voice interaction:** extending the existing browser architecture, which already supports natural-language commands, to full voice-based engineering interaction using standard browser speech APIs.

Throughout this evolution, the platform’s central design commitment is preserved: AI augments engineering expertise, and the engineer remains in control of every decision.

VII. CONCLUSION

This paper presented RANOpt AI, a browser-based, AI-assisted interactive decision-support platform for 5G/6G RAN engineering. By combining deterministic, standards-aligned wireless engineering models with an explainable AI Decision Engine – combining rule-based logic with a PPO-based reinforcement-learning policy already deployed for scheduler-optimization inference, with continued offline pretraining as ongoing work – the platform enables engineers to rapidly evaluate alternative radio configurations while retaining full transparency and final decision authority. A layered validation framework, comprising automated consistency checks, golden engineering scenarios, and diagnostic trace mode, supports reproducible and auditable use of the platform. The current implementation is intentionally scoped to a representative single cell; multi-cell modeling and an AI-native Network Digital Twin, including a planned future upgrade toward an area/morphology-based digital twin using carrier or synthetic

data, represent the next stages of the platform’s evolution toward comprehensive 5G-Advanced and 6G RAN decision support.

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